AVYSCAN LAB

A V Y S C A N L A B

ScanFilm SNC — www.scanfilm.ch

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1. OVERVIEW

Avyscan Lab is a restoration software for digitized film (8 mm, Super 8, 9.5 mm, 16 mm). It allows you to:

- Reduce film grain (MVTools2 MDegrain)
- Remove dust and scratches (RemoveDirt / RemoveDirtMC)
- Correct brightness, contrast, saturation and colors
- Correct color drift (GamMac)
- Crop the image
- Enhance sharpness (edge-aware unsharp mask)
- Stabilize 8 mm film (DePan / DePanEstimate)
- Auto white balance (ColorYUV autowhite)
- RGB levels correction in 32-bit float precision
- White balance in 32-bit float precision (per-channel gain/gamma)
- Add custom AviSynth filters with a full control interface

The software generates an optimized AviSynth+ script and provides real-time preview through an embedded video player (libmpv). Final encoding is handled by FFmpeg.

The application is available in 4 languages: French, English, German and Spanish.

ACKNOWLEDGMENTS / CREDITS

Avyscan Lab relies on the quality of AviSynth+ plugins developed and maintained by the open-source community. We extend our warmest thanks, as exhaustively as possible, to their authors:

MVTools2 — Manao, Fizick, Tsp, TSchniede, SEt
Motion estimation and temporal grain reduction (MDegrain).
The core of Avyscan Lab's restoration pipeline.

RemoveDirt — Kassandro
Temporal dust and scratch detection and removal.
RemoveDirtMC (with motion compensation) remains unmatched for cleaning film stock.

GamMac — StainlessS
Advanced color correction through RGB channel analysis.
A valuable tool for correcting color drift in aging film stock.

MaskTools2 — Manao, tp7
Mask operations, essential for selective sharpness enhancement (edge-aware mode).

RgTools — tp7, pinterf
Grain removal tools and matrix operations, used as a prefilter in motion analysis.

LimitedSharpenFaster — Didée
Sharpness enhancement technique using Sobel mask and gamma curve (^0.86), which inspired Avyscan Lab's "edge" mode.

DePan / DePanEstimate — Fizick
Global motion estimation and stabilization filters.
Used for 8 mm film stabilization, correcting gate weave and frame jitter typical of amateur film projectors.

2. REQUIREMENTS AND INSTALLATION

System requirements:
- Windows 10 or 11 (64-bit)
- 4 GB RAM minimum (8 GB recommended)
- Sufficient disk space for source files and renders

Components:
- AviSynth+ 64-bit must be installed. If it is missing, Avyscan Lab detects it at startup and offers to install it.
- FFmpeg is included with the application (Plugins/ffmpeg/ folder).
- The mpv player is included with the application (mpv/ folder).

No additional installation is required. Simply extract the archive and run the AvyscanLab.exe executable.

3. GENERAL INTERFACE

The interface is organized into three main areas:

UPPER AREA

Menu bar:
- Infos User guide, Script preview, Guided tour, Check for updates
- Settings Threads, Source loader, Theme, Accent color, Export/Import settings, Reset settings
- Languages FR / EN / DE / ES
- Contact About, Feedback form

Source bar: source file selection fields, image/video mode, FPS, threads, source loader (Auto / FFMS2 / L-SMASH / LWLibav).

Clip tabs: each loaded clip appears as a tab with the file name and the applied preset (e.g.: "[perso2]"). Click a tab to switch between clips. The + button adds a clip, the x closes a tab.

Transport bar (below the player):
- |< Beginning of clip
- << Previous frame
- > Play / Pause
- >> Next frame
- >| End of clip
- 1x Speed (cycle: 0.25x -> 0.5x -> 1x)
- 1/2 Half resolution (faster preview)
- F11 Maximize/Restore player
- Hist Histogram (right-click to change type)
- Preset Preset dropdown for the active clip
- REC Open encoding panel

CENTRAL AREA

mpv video player: displays the AviSynth script preview in real time.

Progress bar: current position / total duration.
Click or drag to navigate.

LOWER AREA

Filter panel divided into two columns separated by a resizable splitter:
- Left column: filter activation buttons (ON/OFF, green = active), expand button (>) to show parameters, edit button (pencil) for custom filters. Active and inactive filter sections. Drag-and-drop to reorder filters.
- Right column: detailed parameters of the selected filter, with a reset button (circular arrow) to restore defaults per filter. Drag-and-drop to reorder parameter panels.

The height between the player and the filter panel is adjustable via the horizontal splitter.

4. SOURCE SELECTION

Video file

Drag and drop a video file onto the playback area, or click the button next to the "source" field to open a file browser.
Supported formats: .avi .mp4 .mov .mkv .wmv .webm .m4v .mpeg .mpg

Image sequence

Select a directory containing image files (for example: 00001.tif, 00002.tif, etc.).
Avyscan Lab automatically detects the first and last image of the sequence.
Supported formats: .tif .tiff .jpg .jpeg .png .bmp

Frame rate (FPS)

The "fps" field defines the playback frame rate.
Common values for standard formats:

8 mm: 16 or 18 fps
Super 8: 18 or 24 fps
9.5 mm: 16 fps
16 mm: 18 or 24 fps

Source loader

The drop-down menu allows you to force the decoding plugin:

Auto Automatic detection (recommended)
FFMS2 FFmpegSource2 — maximum compatibility
L-SMASH LSmashSource — optimized for MP4/MOV
LWLibav LWLibavVideoSource — alternative format

If a reading error occurs when opening a file, try changing the source loader.

Threads

Number of AviSynth processing threads (default: 4). Increase this value to use all processor cores.
A value that is too high may cause slowdowns or errors on less powerful machines.
Adjust it according to your hardware.

Multi-clips

You can load multiple files. Each clip has its own filter settings and its own preset. Drag and drop multiple files at once or use the + button to add more.

5. VIDEO PLAYER AND NAVIGATION

The built-in player displays the AviSynth script result in real time. Any parameter change automatically regenerates the script and refreshes the preview.

Keyboard shortcuts:

Key	Action
Space	Play / Pause
Left arrow	Previous frame
Right arrow	Next frame
Ctrl + Left	Go to beginning
Ctrl + Right	Go to end
F11	Maximize/Restore player size

Parameter interactions:

Action	Effect
Click on slider	Positions the cursor directly
Drag slider	Real-time adjustment
Scroll wheel on text field	Adjusts value by step
Direct input	Type a value, confirm with Enter or click elsewhere
Hover over a label	Displays tooltip with range and tips for the selected parameter

Half-resolution mode:
The "1/2" button activates half-resolution mode: the AviSynth script processes the image at half its size BEFORE the heavy filters, then displays it. This significantly speeds up the preview on less powerful machines.

Error display:
In case of an AviSynth script error, a banner appears in the player and an error window opens with:
- The error message (without ffmpeg technical prefixes)
- The script excerpt around the faulty line (with marker >)
- The full ffmpeg output in a collapsible "Details" section
- A "Copy" button to copy the complete message

Per-clip preset:
The drop-down menu next to the transport bar allows you to apply a filter preset to the active clip only.

Histogram:
The histogram button displays a real-time luminance/color histogram overlay on the video player. Right-click the button to cycle through histogram types.

6. FILTER PANEL

The lower panel is divided into two columns:
Left column: activation buttons (ON/OFF, green = active)
Right column: detailed parameters of the selected filter

Each filter has:
- An ON/OFF button to enable/disable the filter
- An expand button (>) that displays its panel with detailed parameters
- An edit button (pencil) for custom/built-in filters to open the editor
- A reset button (circular arrow) to restore default values
- Quick presets (for some filters)

General options (outside filters) are:

preview	AviSynth preview (before/after comparison)
enable_flip_h	Horizontal flip
enable_flip_v	Vertical flip

Processing pipeline order:
1. Stabilisation (8 mm — DePan, before all other processing)
2. Flip (horizontal / vertical)
3. Crop
4. Dust removal (denoise / RemoveDirt)
5. Grain reduction (degrain / MVTools2)
6. Brightness levels (luma levels)
7. RGB levels (32-bit)
8. Color correction (GamMac / Auto white balance / White balance 32-bit)
9. Custom filters (6 possible injection points)
10. Sharpness (sharpen) — always last

Active and inactive sections:
Filters that are toggled ON appear in the ACTIVE section at the top of the left column. Disabled filters move to the INACTIVE section below. This separation provides a clear overview of which processing is applied.

Drag-and-drop reordering:
Filters in the left column can be reordered by dragging them.
Parameter panels in the right column can also be reordered by dragging.
Use the reset order button to restore the default filter arrangement.

Slider values are automatically snapped to the step defined for each parameter, which prevents invalid values.

7. CROP FILTER

Removes pixels from the image edges. Values are automatically rounded to even numbers (YUV chroma subsampling requirement).

Parameter	Range	Default	Description
Crop_L	0-500 px	0	Pixels removed on the left
Crop_T	0-500 px	0	Pixels removed at the top
Crop_R	0-500 px	0	Pixels removed on the right
Crop_B	0-500 px	0	Pixels removed at the bottom

Region draw:

Click the "Draw region" button to define the crop area directly on the video player by drawing a rectangle. The crop values are automatically calculated from the drawn region.

8. RESTORATION FILTERS

DEGRAIN — Grain reduction (MVTools2)
Temporal film grain reduction via MVTools2 motion analysis.
The filter compares multiple consecutive frames to distinguish detail from grain.

Parameter	Range	Default	Description
degrain_mode	mdegrain1/2/3/temporal	mdegrain2	Algorithm
degrain_thSAD	0-1000	350	Luma threshold
degrain_thSADC	0-1000	250	Chroma threshold
degrain_blksize	4/8/16/32/64	16	Block size
degrain_overlap	0 to blksize/2	blksize/2	Overlap
degrain_pel	1 or 2	1	Sub-pixel precision
degrain_search	0-5	2	Search mode
degrain_prefilter	remgrain/blur/none	remgrain	Prefilter

Automatic constraints:
- blksize can only take values 4, 8, 16, 32 or 64 (powers of 2, MVTools2 MAnalyse requirement). The slider adjusts automatically.
- overlap is automatically capped at blksize/2. If you reduce blksize, overlap is reduced accordingly.
- overlap only accepts even values (step = 2).

Quick presets:

Light	mdegrain1, thSAD=150, thSADC=100
Standard	mdegrain2, thSAD=350, thSADC=250 (default)
Medium	mdegrain2, thSAD=550, thSADC=380
Strong	mdegrain3, thSAD=750, thSADC=520
Very strong	mdegrain3, thSAD=1000, thSADC=700

Notes:
- mdegrain2: 2 temporal references (recommended, good quality/speed ratio)
- mdegrain3: 3 references (better result, slower)
- mdegrain1: 1 reference (fast, less effective)
- temporal : TemporalSoften (without MVTools2, basic)
- prefilter: smoothing applied to a temporary copy of the clip to help MVTools estimate motion vectors. Does NOT filter the output image. remgrain = RemoveGrain(2) fast (recommended). blur = Blur(1.0). none = no prefiltering.

DENOISE — Dust removal (RemoveDirt)
Temporal detection and removal of dust and spot defects.

Parameter	Range	Default	Description
denoise_mode	removedirtmc/removedirt	removedirtmc	Algorithm
denoise_strength	1-24	10	Detection threshold
denoise_dist	1-20	3	Spatial radius
denoise_grey	true/false	false	Luma only

Quick presets:

Light	strength=4, dist=2
Standard	strength=10, dist=3 (default)
Medium	strength=14, dist=4
Strong	strength=18, dist=6
Very strong	strength=24, dist=10

Note: removedirtmc (with motion compensation) is recommended for better results. The removedirt mode is faster but less accurate.

8MM STABILISATION — DePan
Image stabilization for 8 mm film using DePan and DePanEstimate plugins.

Corrects gate weave and frame jitter typical of amateur film projectors and telecine machines. Placed at the very beginning of the pipeline (BeforePipeline), before all other filters.

Parameter	Range	Default	Description
maxstabH	0-30 %	14	Max horizontal range (% of width)
maxstabV	0-30 %	14	Max vertical range (% of height)
error	1-200	65	Motion estimation error threshold
mirror	0/1	0	Border fill (0=black, 1=mirror edges)
cutoff	0.0-5.0	0.5	High-pass cutoff frequency
damping	0.0-5.0	1.0	Damping of stabilization response
est_left	0-200 px	40	Estimation zone: ignore left pixels
est_top	0-200 px	40	Estimation zone: ignore top pixels
est_right	0-200 px	40	Estimation zone: ignore right pixels
est_bottom	0-200 px	40	Estimation zone: ignore bottom pixels
crop_left	0-200 px	56	Final crop left after stabilization
crop_top	0-200 px	40	Final crop top after stabilization
crop_right	0-200 px	56	Final crop right after stabilization
crop_bottom	0-200 px	40	Final crop bottom after stabilization

Notes:

- The estimation zone (est_*) defines the area used for motion analysis. Exclude film sprocket holes and borders by increasing these values.
- The final crop (crop_*) removes the black borders introduced by the stabilization shift. Set these values larger than maxstabH/V percentages to avoid visible black edges.
- mirror=1 fills borders with mirrored edges instead of black, hiding the stabilization crop effect.
- cutoff controls the frequency separation: lower values remove slow drift (panning), higher values only correct fast jitter.
- damping controls how smoothly the stabilization responds; higher values give smoother but slower correction.

Region draw:

Click "Draw region" to define the estimation zone directly on the video player. The est_left/top/right/bottom values are automatically set from the drawn rectangle.

9. ENHANCEMENT FILTERS

LUMA LEVELS — Brightness levels
Global adjustment of brightness, contrast, saturation and hue.

Parameter	Range	Default	Description
Lum_Bright	-255 to 255	0	Brightness offset
Lum_Contrast	0.1 to 3.0	1.05	Contrast multiplier
Lum_Sat	0.1 to 3.0	1.10	Color saturation
Lum_Hue	-180 to 180	0	Hue rotation (degrees)
Lum_GammaY	0.1 to 3.0	1.30	Gamma curve

Note: GammaY > 1 brightens midtones, < 1 darkens them.
This is often the most useful parameter for underexposed old films.

RGB LEVELS (32-BIT) — Precision color grading
Per-channel RGB correction in 32-bit floating-point precision.
Provides fine-grained control over brightness, contrast, gamma,
and individual channel gain/gamma/offset for professional color grading.

Parameter	Range	Default	Description
brightness	-0.3 to 0.3	0.0	Global brightness offset
contrast	0.5 to 2.0	1.0	Global contrast multiplier
gamma	0.2 to 5.0	1.0	Global gamma (<1 darker, >1 brighter)
r_gain	0.5 to 2.0	1.0	Red channel gain (multiplier)
r_gamma	0.2 to 5.0	1.0	Red channel gamma
r_offset	-0.2 to 0.2	0.0	Red channel offset (lift)
g_gain	0.5 to 2.0	1.0	Green channel gain
g_gamma	0.2 to 5.0	1.0	Green channel gamma
g_offset	-0.2 to 0.2	0.0	Green channel offset (lift)
b_gain	0.5 to 2.0	1.0	Blue channel gain
b_gamma	0.2 to 5.0	1.0	Blue channel gamma
b_offset	-0.2 to 0.2	0.0	Blue channel offset (lift)

Notes:
- Processing is done in 32-bit float for maximum precision, avoiding banding artifacts that can occur with 8-bit corrections.
- Use per-channel gain to correct color casts: increase r_gain to warm the image, increase b_gain to cool it.
- Per-channel offset (lift) shifts the black point of each channel independently.
- Per-channel gamma adjusts midtones without clipping highlights.

GAMMAC — Advanced color correction
Automatic color drift correction between RGB channels. Particularly
useful for films showing yellowing, blue or red color casts.

Parameter	Range	Default	Description
LockChan	-3 to 2	1	Reference channel
LockVal	1-255	250	Lock threshold
Scale	0-2	2	Scale mode
Th	0.0-1.0	0.12	Low threshold
HiTh	0.0-1.0	0.25	High threshold
X	0-2000	0	Analysis area X offset
Y	0-2000	0	Analysis area Y offset
W	0-4000	0	Area width (0 = all)
H	0-4000	0	Area height (0 = all)
Omin	0-255	0	Minimum output
Omax	0-255	255	Maximum output
Show	true/false	false	Display visualization

Verbosity	0-6	4	Log detail level

LockChan: 0=Red, 1=Green, 2=Blue, -1=target value, -2=average, -3=median.
Click the (i) button to open the detailed GamMac documentation.

Region draw:
Click "Draw region" to define the GamMac analysis area (X/Y/W/H) directly on the video player. The values are automatically set from the drawn rectangle.

AUTO WHITE BALANCE — ColorYUV autowhite
Automatic white balance correction using AviSynth's built-in ColorYUV autowhite function. Adjusts color channels to achieve neutral whites.

Parameter	Range	Default	Description
strength	0.0 to 1.5	1.0	Blend strength (0=none, 1=full)
autogain	true/false	false	Also auto-adjust brightness levels

Notes:
- Strength controls the blend between the original and the corrected image. Use values below 1.0 for a subtle correction, above 1.0 for an exaggerated effect.
- The autogain checkbox enables automatic brightness level adjustment in addition to white balance.
- Placed after GamMac in the pipeline; can be combined with GamMac or used as an alternative.

WHITE BALANCE (32-BIT) — Per-channel correction
Manual white balance correction in 32-bit floating-point precision.
Provides per-channel gain and gamma for precise color cast removal.

Parameter	Range	Default	Description
r_gain	0.5 to 2.0	1.0	Red gain (>1 warms, <1 cools)
r_gamma	0.2 to 3.0	1.0	Red gamma (midtone adjustment)
g_gain	0.5 to 2.0	1.0	Green gain (reference channel)
g_gamma	0.2 to 3.0	1.0	Green gamma
b_gain	0.5 to 2.0	1.0	Blue gain (>1 cools, <1 warms)
b_gamma	0.2 to 3.0	1.0	Blue gamma

Notes:
- Processing is done in 32-bit float for maximum precision.
- Gain is an overall channel multiplier (1.0 = neutral).
- Gamma adjusts midtones without clipping highlights: >1 brightens the channel midtones, <1 darkens them.
- Placed after GamMac in the pipeline.

SHARPEN — Sharpness enhancement
Image sharpness enhancement. Always applied last in the pipeline to avoid amplifying artifacts from other filters.

Parameter	Range	Default	Description
Sharp_Mode	simple/edge	simple	Algorithm
Sharp_Strength	1-20	8	Enhancement strength
Sharp_Radius	0.5-5.0	1.5	Blur radius (mask)
Sharp_Threshold	0-100	5	Anti-noise threshold

Quick presets:

Light	simple, Strength=5, Radius=1.0
Standard	simple, Strength=8, Radius=1.2 (default)
Medium	edge, Strength=12, Radius=1.5
Strong	edge, Strength=16, Radius=1.8, Threshold=20
Very strong	edge, Strength=20, Radius=2.5, Threshold=35

Notes:
- simple: global unsharp mask, effective but may amplify residual grain.

- edge: edges only via dual Sobel mask (horizontal + vertical) with gamma compression (^0.86, LimitedSharpenFaster / Didée technique). Preserves smooth areas. Recommended after Degrain.
- Radius controls the extent of the sharpening halo: 1.0 = tight, precise; 3.0 = broad, pronounced.
- Threshold excludes weak responses from the edge mask. Recommended: 20-40 for noisy footage. Set to 0 to disable (gamma compression alone is usually sufficient).

10. CUSTOM FILTERS

Avyscan Lab allows you to create your own AviSynth filters and integrate them into the restoration pipeline with a full control interface.

CREATING A CUSTOM FILTER

1. Click the "+" button to the right of the "CUSTOM" title in the filter panel.
   The custom filter form opens.

2. Enter the filter name and its position in the pipeline:
   - BeforePipeline : before all filters
   - AfterDenoise : after dust removal
   - AfterDegrain : after grain reduction
   - AfterLuma : after brightness levels (Luma / RGB Levels)
   - AfterGamMac : after color correction (GamMac / AWB / WB)
   - AfterSharpen : after sharpness (end of pipeline)

3. If your filter requires an external AviSynth plugin (.dll), click
   "+ Add DLL..." to add it. DLLs are loaded via TryLoadPlugin()
   and duplicates are automatically eliminated.

4. Write the AviSynth code in the editor. Use "c" as the clip variable:
   c = MyFilter(c, strength={sigma})
   Words enclosed in braces {name} are automatically detected as parameters.

   Tip: select a line like "var = value" with the mouse — a prompt offers
   to convert it into a {placeholder} control automatically.

5. For each detected parameter, configure the control type:
   - slider : slider with Min, Max, Step and Default.
     Values are automatically snapped to the step.
     With a step of 2, only even values are allowed.
   - combo : drop-down list with comma-separated values
   - checkbox : check box with customizable On/Off values
   - text : free text field

6. Per-parameter buttons:
   - Eye icon: highlights the {placeholder} in the code editor and scrolls to it
   - Red X: removes the control and replaces {placeholder} with its default
     value in the code

7. Use the "Preview" button to test the filter in real time without leaving
   the form.

8. Click "Save" to save. Other footer buttons:
   - Delete: permanently deletes this custom filter
   - Duplicate: creates an independent copy with a new name
   - Export: saves the filter as a JSON file
   - Cancel: closes without saving

REGION DRAW SUPPORT

Custom filters that use X/Y/W/H parameters can enable the "Region draw" checkbox. When enabled, a "Draw region" button appears in the filter panel, allowing the user to define the analysis or effect area directly on the video player.

USING A CUSTOM FILTER

The filter appears in the CUSTOM section with:
- An ON/OFF button to enable/disable the filter
- An arrow button (>) to open the parameter panel
- A pencil button to open the edit form

Parameters are displayed in the right column of the filter panel with the defined controls (sliders, lists, check boxes, text fields).

EXPORTING / IMPORTING A FILTER

Export: in the edit form, click "Export". The filter is saved to a .json file that you can share with other users.

Import: in the filter panel, click the down arrow button next to the "+" in the CUSTOM section. Select an exported .json file. Imported filters receive a new identifier to avoid duplicates.

Note: DLL paths are absolute. If the filter uses an external plugin, make sure the DLL is present at the same location on the target machine, or edit the filter after import to correct the path.

DUPLICATING A FILTER

Open the edit form (pencil button) and click "Duplicate". An independent copy is created with a new name and a new identifier. Useful for creating variations of the same filter with different default values.

DELETING A FILTER

Open the edit form (pencil button) and click "Delete".
Built-in filters that are removed will be restored on next launch.

Storage location: %AppData%\AvyscanLab\custom_filters.json

IMPORTING AN AVSI SCRIPT

Some AviSynth filters rely on functions defined in script files (.avsi). These files contain reusable functions that extend AviSynth's capabilities without being compiled plugins (.dll).

To add an .avsi script to a custom filter:

1. Open the filter edit form (pencil button or create a new filter via the "+" button).
2. In the "Scripts (Import .avsi)" section, click "+ Add AVSI...".
3. Select one or more .avsi or .avs files from your disk.
   The selected files are automatically copied to the application's Plugins/Scripts/ folder, which ensures their availability regardless of their original location.
4. The added scripts appear in the list. Use the (x) button to remove a script from the list.
5. When generating the AviSynth script, Avyscan Lab automatically inserts

an Import() instruction for each referenced script, right after loading the plugins:

```
TryLoadPlugin("C:/.../MyPlugin.dll")
Import("C:/.../Plugins/Scripts/MyFunctions.avsi")
```

The functions defined in the .avsi file are then available in the code of your custom filter.

Example: if your file MyFunctions.avsi defines a function MyDenoise(), you can use it directly in the filter code:

```
c = MyDenoise(c, strength={sigma})
```

Note: when exporting a filter, only the .avsi file name is saved (not its contents). If you share a filter with another user, also provide the necessary .avsi files. The user will need to import them via the same mechanism.

11. PRESETS

GLOBAL PRESETS (filters)
The Infos > Preset menu allows you to save and restore your filter configurations.

Action	Description
Save	Saves the current settings under a name
Load	Applies an existing preset
Update	Overwrites an existing preset with the current settings
Delete	Removes a preset

When applying a preset, you can choose to apply it to the active clip only or to all loaded clips.

The source, film, img, start, end and crop fields are NOT included in presets.
Location: %AppData%\AvyscanLab\presets.json

PER-CLIP PRESETS

Each clip can have its own filter settings. When you manually modify a parameter, the clip is automatically named "perso1", "perso2", etc.
The drop-down menu in the transport bar allows you to load a global preset for the active clip only.

When a preset parameter is modified, a dialog asks whether to save the change to the preset. This can be disabled with "Don't ask me again" to always save automatically.

ENCODING PRESETS

Encoding combinations (encoder + container + quality + chroma + resolution) can be saved and reloaded from the Record panel.
The "Default" preset is always present and appears first.
Location: %AppData%\AvyscanLab\encoding_presets.json

GAMMAC PRESETS

GamMac color correction settings can be saved and loaded as separate presets.
This allows you to store frequently used color correction profiles (e.g., for specific film stocks or aging patterns) and quickly apply them.
Location: %AppData%\AvyscanLab\gammac_presets.json

12. ENCODING AND RENDERING

Click the REC button (red) in the transport bar to open the encoding panel.

STEP 1 — Clip selection
Check the clips to encode in the list. Use "Select all" to encode them all at once (batch). Each clip can have a different output name and a different encoding preset.
Each preset can include a different destination folder, making it easy to organize the renders of multiple clips.

STEP 2 — Destination folder
Click the Browse button or enter the path. Available disk space is displayed in real time.

STEP 3 — Encoder selection

Encoder	Type	Usage
x264	Lossy	Distribution (H.264)
x265	Lossy	Distribution (HEVC)
FFV1	Lossless	Professional archiving
UT Video	Lossless	Fast editing
ProRes	Lossy HQ	Post-production
TIFF seq.	Lossless	Individual images (archiving)
PNG seq.	Lossless	Individual images
JPG seq.	Lossy	Individual images

STEP 4 — Container
MKV, MP4, AVI or MOV (for video codecs). Ignored for image sequences.

STEP 5 — Advanced settings (lossy codecs only)
Quality mode:
- CRF Constant quality (default). The lower the value, the better the quality. Typical values: 18-23. The slider ranges from 0 to 51.
- Bitrate Fixed bitrate in Mb/s. The recommended minimum is automatically calculated based on the chosen resolution and chroma.

Chroma subsampling:
- 4:2:0 Standard (smaller file)
- 4:2:2 Better color quality
- 4:4:4 Maximum quality (large file)

Note: bitrate and chroma are coupled. Changing the chroma adjusts the recommended minimum bitrate. Entering a high bitrate may automatically adjust the chroma to the maximum supported level.

Output resolution:
- Original Source resolution preserved
- 1080p 1920x1080
- 720p 1280x720
- 576p 720x576 (PAL)
- 480p 720x480 (NTSC)

Note: odd dimensions are automatically adjusted to even numbers (x264/x265 requirement) via a padding filter.

DURING ENCODING

- Video playback is automatically stopped
- The interface is locked (filters, transport, menus, clip tabs)
- Only the "Stop" button remains active

- A progress bar shows the completion percentage (Clip X/Y for batch)
- Closing the application automatically stops the current encoding

"Shut down after encoding" option: check this box to shut down the computer automatically after a successful batch encoding. A 60-second warning is displayed before shutdown.

File overwrite: if the output file already exists, a confirmation dialog asks whether to overwrite or skip.

Safety check: if the output would overwrite the source file, the operation is blocked with a warning message.

Technical note: encoding uses a dedicated render script (ScriptRender.avs) where preview mode is disabled, ensuring maximum final quality.

Trial version: limited to 3 files, 1 minute of encoding each.

13. APPLICATION SETTINGS

Settings menu:

Parameter	Description
Theme	Dark (default), Grey, or Light
Accent	Blue (default), Green, Red, Violet, Orange
Threads	Number of AviSynth processing threads
Source loader	Decoding plugin (Auto/FFMS2/L-SMASH/LWLibav)
Export settings	Exports all settings (presets, custom filters, encoding presets, window settings, theme) to a ZIP archive for backup or transfer
Import settings	Imports settings from a previously exported ZIP archive. An import report shows the status of each file before applying
Reset settings	Resets all settings to default and restarts. Warning: this is a complete reset that erases all presets, custom filters and other saved configurations

Languages menu:

Immediate language change. The interface, AviSynth scripts and guide are regenerated in the chosen language.
Available: French (FR), English (EN), German (DE), Spanish (ES).

14. SCRIPT PREVIEW

The Infos > Script preview menu opens a window displaying the full contents of the ScriptUser.avs file generated by Avyscan Lab.

The "Reload" button re-reads the file from disk and refreshes the preview.
This allows you to check the AviSynth script produced and optionally copy it for external use (AvsPmod, VirtualDub, etc.).

15. FEEDBACK

The Contact menu opens a built-in contact form allowing you to send directly to the developer:

- A bug report (with writing tips)
- A feature request
- A general review with rating (stars from 1 to 5)

The form is submitted via the Internet. Name and email are optional but recommended if you would like a response.

16. GUIDED TOUR

The Infos > Guided tour menu launches an interactive step-by-step tour of the application interface. The tour covers 7 steps:

1. Welcome — Introduction to Avyscan Lab
2. Add a clip — How to load a video or image sequence
3. Enable filters — Toggling filters and expanding parameters
4. Adjust parameters — Using sliders and real-time preview
5. Preview — Transport controls and playback
6. Record — Opening the encoding panel
7. Choose output and start — Selecting the output directory Launching the encoding

Each step highlights the relevant area of the interface. Navigation buttons (Next, Back, Skip) allow you to move through the tour at your own pace. The tour ends with a "Get started!" button.

A language selector is available during the tour to switch languages.

17. RECOMMENDED WORKFLOW

1. Open your source (drag and drop or source field).
2. Navigate to a representative frame (neither too dark nor too bright).
3. Enable the filters one by one in pipeline order:

   a) Stabilisation — For 8 mm film: enable the 8mm Stabilisation filter to correct gate weave. Adjust the estimation zone to exclude sprocket holes and borders.

   b) Crop — Remove black borders or adjust framing.

   c) Denoise — Remove dust and scratches. The "standard" preset works in most cases. Increase strength if defects persist.

   d) Degrain — Start with the "Standard" preset, then increase thSAD if grain persists. Warning: values that are too high produce a "plastic" effect.

   e) Luma — Adjust brightness and contrast. GammaY is often the most useful parameter for brightening old films.

   f) RGB Levels — For precise color grading in 32-bit: adjust per-channel gain, gamma and offset.

   g) GamMac — Correct color drift (yellowing, blue cast, red cast, etc.).

   h) AWB / WB — Use Auto White Balance for automatic correction, or White Balance 32-bit for manual per-channel gain/gamma control.

   i) Sharpen — Enhance sharpness as the last step. The "edge" mode better preserves smooth areas and avoids amplifying residual grain.

4. Use the quick presets available on some filters for fast adjustment, then fine-tune.
5. Verify the result by browsing through several passages of the film.
6. Save your configuration as a preset in the Infos menu so you can easily reuse it on other clips from the same film or similar films.
7. If you have multiple clips: load them all, then apply the appropriate presets.
8. Open the Record panel, choose the encoding settings.
   Each film can have a different encoding preset, making it easy to organize the renders.
   Each film can be included or not in a batch encoding.
   Start the render!

Tip: enable half-resolution mode (1/2) while adjusting parameters to speed up the preview, then disable it to verify the result at full resolution before encoding.

18. TIPS AND SHORTCUTS

- The mouse scroll wheel on numeric fields adjusts the value by step.
- Half-resolution mode (1/2 button) significantly speeds up the preview.
- All parameters are automatically saved to ScriptUser.avs with each change.
- The window position, size and panel states are remembered between sessions.
- The Record panel is restored on startup if it was open when closing.
- Tooltips (on hover) on each parameter display ranges and tips.
- Dragging and dropping multiple files automatically creates one clip per file.
- Sliders position directly on click (no need to drag).
- The guided tour (Infos menu) lets you discover the interface basics step by step.
- AviSynth errors automatically display the faulty script line.
- The "Copy" button in the error window copies the message and details.
- Drag-and-drop filters in the left column to reorder them.
- Drag-and-drop parameter panels in the right column to reorder them.
- Use the reset order button to restore the default filter arrangement.
- The histogram button shows a real-time overlay; right-click to change type.
- Custom filters can be duplicated to create variations with different defaults.
- Export/Import settings via the Settings menu for backup or transfer to another machine.

19. FILES AND LOCATIONS

File / Folder	Location
ScriptUser.avs	Exe folder
ScriptMaster.xx.avs	Exe folder (xx = language)
ScriptRender.avs	Exe folder (generated for encoding)
presets.json	%AppData%\AvyscanLab
encoding_presets.json	%AppData%\AvyscanLab
gammac_presets.json	%AppData%\AvyscanLab
custom_filters.json	%AppData%\AvyscanLab
window-settings.json	%AppData%\AvyscanLab
session.json	%AppData%\AvyscanLab
theme-settings.json	%AppData%\AvyscanLab
cfdlg-layout.json	%AppData%\AvyscanLab
Plugins/	Exe folder
mpv/	Exe folder
Filters/	Exe folder (built-in filter JSON definitions)
Users Guide/	Exe folder

20. APPLICATION CONTENTS

File tree distributed with Avyscan Lab:

```
AvyscanLab.exe                    Main executable
ScriptMaster.en.avs               AviSynth script template

Assets/
  Ico.ico                         Application icon
  Logo.png                        Logo

mpv/
  libmpv-2.dll                    Video player (libmpv)

Users Guide/
  AvyscanLab_Guide_en.txt         User guide (English)
  AvyscanLab_Guide_fr.txt         User guide (French)
  AvyscanLab_Guide_de.txt         User guide (German)
  AvyscanLab_Guide_es.txt         User guide (Spanish)

Filters/
  stab8mm.json                    8mm Stabilisation (built-in)
  rgb_levels32.json               RGB Levels 32-bit (built-in)
  awb32.json                      Auto White Balance (built-in)
  wb32.json                       White Balance 32-bit (built-in)
  degrain.json                    Degrain (built-in)
  denoise.json                    Denoise (built-in)
  luma.json                       Luma Levels (built-in)
  sharpen.json                    Sharpen (built-in)
  gammac.json                     GamMac (built-in)
  crop.json                       Crop (built-in)
  flip_h.json                     Horizontal flip (built-in)
  flip_v.json                     Vertical flip (built-in)

Plugins/
  Avisynth/
    AviSynth.dll                  Bundled AviSynth+ engine
    DevIL.dll                     Image loading (dependency)
    ImageSeq.dll                  Image sequence support

  Dual/L-Smash-Works/
    LSMASHSource.dll              L-SMASH decoder (MP4/MOV)
    LSMASHSource_indexing.exe     Indexing tool

  ffmpeg/
    ffmpeg.exe                    Video encoder
    ffprobe.exe                   Video file analysis
    avcodec-62.dll                FFmpeg codec
    avdevice-62.dll               FFmpeg devices
    avfilter-11.dll               FFmpeg filters
    avformat-62.dll               FFmpeg formats
    avutil-60.dll                 FFmpeg utilities
    swresample-6.dll              Audio resampling
    swscale-9.dll                 Video scaling

  FFMS2/
    ffms2.dll                     FFMS2 decoder

  GamMac/
    GamMac_x64.dll                GamMac color correction

  Masktools2/
```

masktools2.dll	Mask operations
MVTools2/	
mvtools2.dll	Motion estimation and compensation
DePan.dll	DePan stabilization
DePanEstimate.dll	DePan estimation
RemoveDirt/	
RemoveDirt.dll	Dust removal
RgTools/	
RgTools.dll	Grain removal tools
Scripts/	
FFMS2.avsi	FFMS2 import script
RemoveDirt.avsi	RemoveDirt import script

AVYSCAN LAB

ScanFilm SNC
www.scanfilm.ch

Support: use the built-in Feedback form in the application.